Exercises for Chapter 3 (second part), with solutions

Graph Algorithms

UP FAMNIT, 2025/2026

November 12, 2025

The exercises cover some of the material covered in Chapter 3. Each exercise is either suggested or more challenging. Suggested exercises deal with concepts from lectures and are similar to exercises that you may expect on exams. More challenging exercises address concepts from lectures, but these exercises are less likely to appear on exam or are intended for a higher grade.

1. *(Decision problems and their complexity; suggested.)*

   Show that the following problem is NP-complete.

Covering Nonedges	
Input:	A graph $G = (V, E)$, integer k.
Question:	Is there a set $S \subseteq V$ such that $\|S\| \leq k$ and for every two distinct nonadjacent vertices x and y of the graph G, at least one of x and y belongs to S?

   Solution:

   The problem is clearly in NP: given a set $S \subseteq V$, we can check in polynomial time if $|S| \leq k$ and for every two distinct nonadjacent vertices x and y of the graph G, at least one of x and y belongs to S.

   To prove NP-completeness, we make a reduction from Vertex Cover. We map an input instance $(G = (V, E), k)$ for Vertex Cover to an input $(\overline{G}, k)$ for Covering Nonedges. Then, for every set $S \subseteq V$ it holds that S is a vertex cover in the graph G with cardinality at most k if and only if $|S| \leq k$ and for every two distinct non-adjacent vertices x and y of the graph $\overline{G}$ at least one of x and y belongs to S.

2. *(Polynomial equivalence of problems; suggested.)*

   Show that there is a Cook reduction between any two of the following problems. In other words, show that the existence of a polynomial-time algorithm for one of them implies the existence of a polynomial-time algorithm for all the others.

   - Maximum Independent Set: For a given graph G, compute $\alpha(G)$, the maximum size of an independent set in G.
   - Minimum Vertex Cover: For a given graph G, compute $\tau(G)$, the minimum size of a vertex cover in G.
   - Maximum Clique: For a given graph G, compute $\omega(G)$, the maximum size of a clique in G.
   - Clique: Given a graph G and a positive integer k, does G have a clique of cardinality at least k?

   Solution: For every graph G it holds that $\alpha(G) + \tau(G) = |V(G)|$, $\alpha(G) = \omega(\overline{G})$, and and $\omega(G) = \alpha(\overline{G})$. With the help of these relations, we can easily establish polynomial equivalence of the problems Maximum Independent Set, Minimum Vertex Cover and Maximum Clique, as follows.

- Assume that A_α is a polynomial-time algorithm for the MAXIMUM INDEPENDENT SET problem. Then we can solve theMINIMUM VERTEX COVER problem in polynomial time as follows: for an input graph G we compute $\alpha(G)$ with the algorithm A_α and return $\tau(G) = |V(G)| - \alpha(G)$.

  Similarly, if A_τ is a polynomial-time algorithm for the MINIMUM VERTEX COVER problem, then we can solve the MAXIMUM INDEPENDENT SET problem in polynomial time: for an input graph G we compute $\tau(G)$ with the algorithm A_τ and return $\alpha(G) = |V(G)| - \tau(G)$.

  So the MAXIMUM INDEPENDENT SET and MINIMUM VERTEX COVER problems are polynomially equivalent.

- Assume that A_α is a polynomial-time algorithm for the MAXIMUM INDEPENDENT SET problem. Then we can solve the MAXIMUM CLIQUE problem in polynomial time, as follows: for an input graph G we compute its complement $\overline{G}$, with the algorithm A_α we compute $\alpha(\overline{G})$, and we return $\omega(G) = \alpha(\overline{G})$.

  Similarly, if A_ω is a polynomial-time algorithm for the MAXIMUM CLIQUE problem, then we can solve the MAXIMUM INDEPENDENT SET problem in polynomial time by computing the complement $\overline{G}$, computing its clique number $\omega(\overline{G})$ with the algorithm A_ω, and returning $\alpha(G) = \omega(\overline{G})$.

  So the MAXIMUM INDEPENDENT SET and MAXIMUM CLIQUE problems are polynomially equivalent.

- A polynomial equivalence of the MINIMUM VERTEX COVER and MAXIMUM CLIQUE follows from transitivity.

It remains to prove the polynomial equivalence of the MAXIMUM CLIQUE and CLIQUE problems.

Assume first that A_ω is a polynomial-time algorithm for the MAXIMUM CLIQUE problem. Then we can solve the CLIQUE problem in polynomial time, as follows: for an input (G, k) we compute with the algorithm A_ω the clique number $\omega(G)$ and check whether $\omega(G) \geq k$ holds. If the inequality holds, then G has a clique of size at least k, otherwise it does not.

Let us now assume that A'_ω is a polynomial-time algorithm for the CLIQUE problem. Then we can solve the MAXIMUM CLIQUE problem in polynomial time, as follows: for an input graph G we use the algorithm A'_ω for all $k \in \{1, \ldots, |V(G)|\}$ to check whether G has a clique of size at least k. The clique number $\omega(G)$ is then the largest $k \in \{1, \ldots, |V(G)|\}$ such that G has a clique of size at least k.

3. *(Polynomial equivalence of problems; more challenging.)*

   Show that the following two problems are polynomially equivalent, i.e., that any one of them is Cook reducible to the other one.

HAMILTONIAN PATH	
Input:	A graph $G = (V, E)$.
Question:	Does the graph G have a Hamiltonian path, i.e., a path on $\|V\|$ vertices?

HAMILTONIAN CYCLE	
Input:	A graph $G = (V, E)$.
Question:	Does the graph G have a Hamiltonian cycle, i.e. a cycle on $\|V\|$ vertices?

Solution: Assume first that A is a polynomial-time algorithm for the Hamiltonian Path problem. Let $G = (V, E)$ be a graph representing the input for the Hamiltonian Cycle problem. For any edge $e = uv$ of graph G, let G'_e be a graph with vertex set $V \cup \{u', v'\}$, where u' and v' are two new vertices, and with edge set $(E \setminus \{e\}) \cup \{uu', vv'\}$. Observe that the graph G has a Hamiltonian cycle containing the edge e if and only if the graph $G - e$ has a Hamiltonian path with one endpoint u and another one v. This last condition is equivalent to the condition that the graph G'_e has a Hamiltonian path (since every Hamiltonian path in the graph G'_e has one endpoint u and another v). So, the graph G has a Hamiltonian cycle if and only if at least one of the graphs G'_e, $e \in E$, has a Hamiltonian path. By computing all the graphs G'_e and using the algorithm A on each of them, we can develop a polynomial-time algorithm that correctly solves the Hamiltonian Cycle problem for a given input graph G.

Let us now assume that A' is a polynomial-time algorithm for the Hamiltonian Cycle problem. Let $G = (V, E)$ be a graph representing the input for the Hamiltonian Path problem. Let G' be a graph with vertex set $V \cup \{x\}$, where x is a new vertex, and the edge set $E \cup \{xv : v \in V\}$. Observe that G has a Hamiltonian path if and only if G' has a Hamiltonian cycle. Ba using the algorithm A' on the graph G', we can develop a polynomial-time algorithm that solves the Hamiltonian Path problem for a given input graph G.[1]

[1] A careful reader will notice that this is a standard (Karp) polynomial reduction. But this is ok, since every Karp reduction is also a Cook reduction. Alternatively, one could obtain a more complicated Cook reduction as follows. For any pair of distinct vertices $s, t \in V$, let $G'_{s,t}$ be the graph with vertex set $V \cup \{x\}$, where x is a new vertex, and the edge set $E \cup \{sx, tx\}$. Note that for arbitrary distinct vertices $s, t \in V$ the following holds: the graph G has a Hamiltonian path with endpoints s and t if and only if it the graph $G'_{s,t}$ has a Hamiltonian cycle. So, a graph G has a Hamiltonian path if and only if at least one of the graphs $G'_{s,t}$ has a Hamiltonian cycle. By computing all the graphs $G'_{s,t}$ and using the algorithm A' on each of them, we can develop a polynomial-time algorithm that solves the Hamiltonian Path problem for a given input graph G.